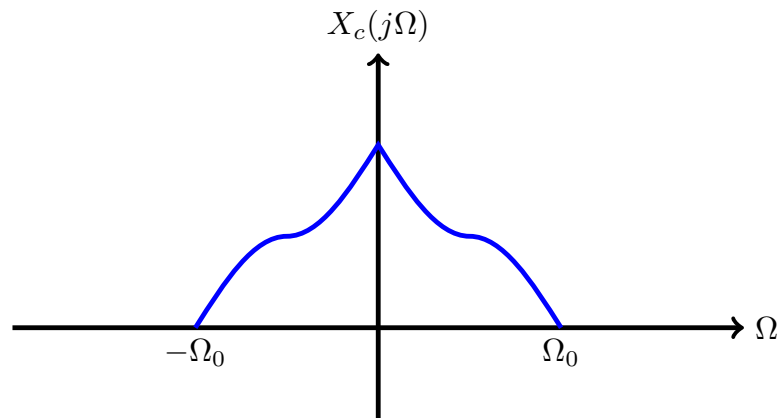


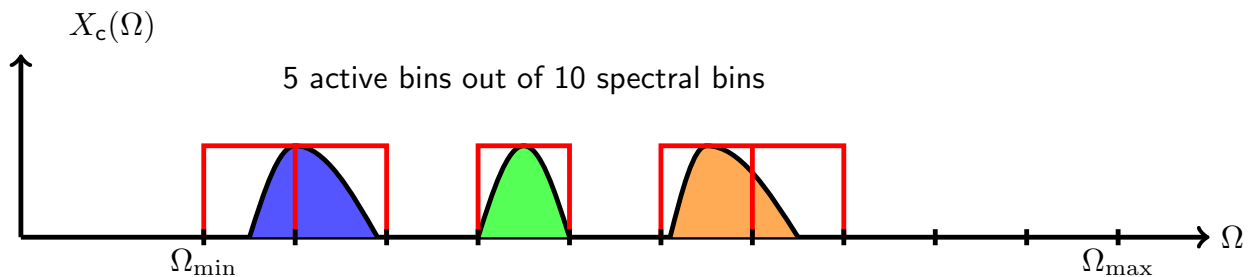
Sub-Nyquist Sampling of Multiband Signals:

- Conventional Nyquist sampling of bandlimited signals



$$x[n] = x_c(nT), \quad \frac{\pi}{T} \geq \Omega_0$$

- Multiband signal (still bandlimited but...)



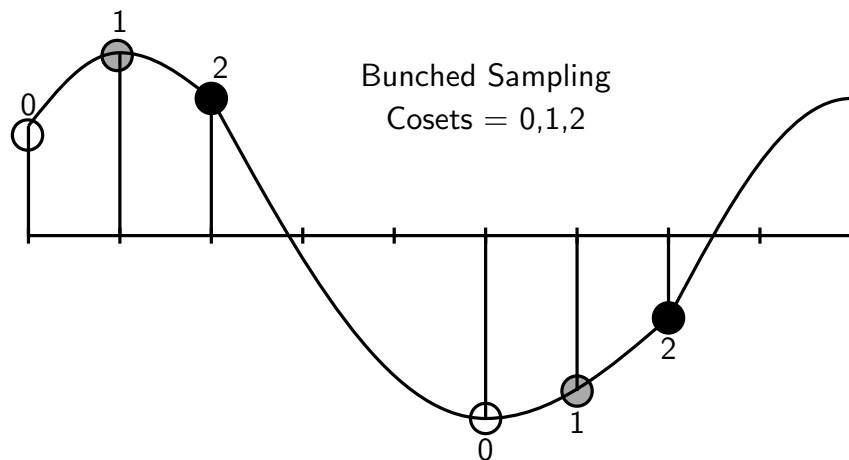
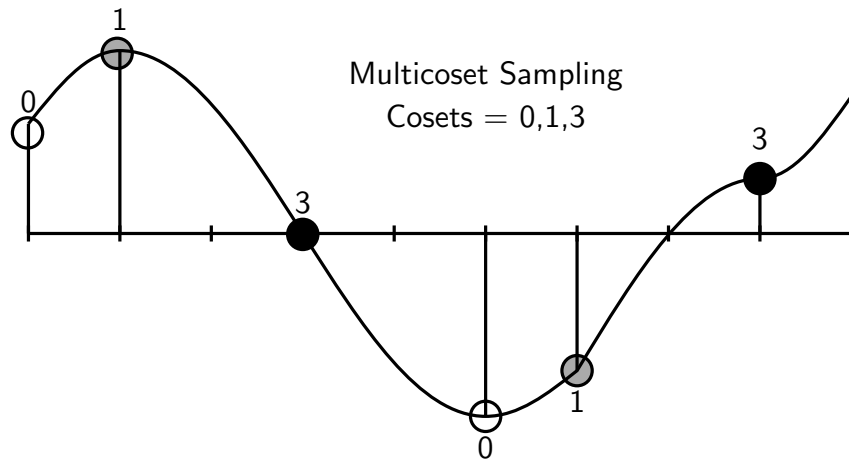
- Q: Can we do better than Nyquist Sampling? How?

Multicoset Sampling:

Let $\Phi = \{\phi_1, \phi_2, \dots, \phi_p\}$ be a subset of $\{0, 1, \dots, L-1\}$ of size $p < L$. We acquire p discrete-time signals by

$$x_{\phi_k}[m] = x_c(LTm + \phi_k T), \quad k = 1, 2, \dots, p.$$

Example: $L = 4, p = 3$,

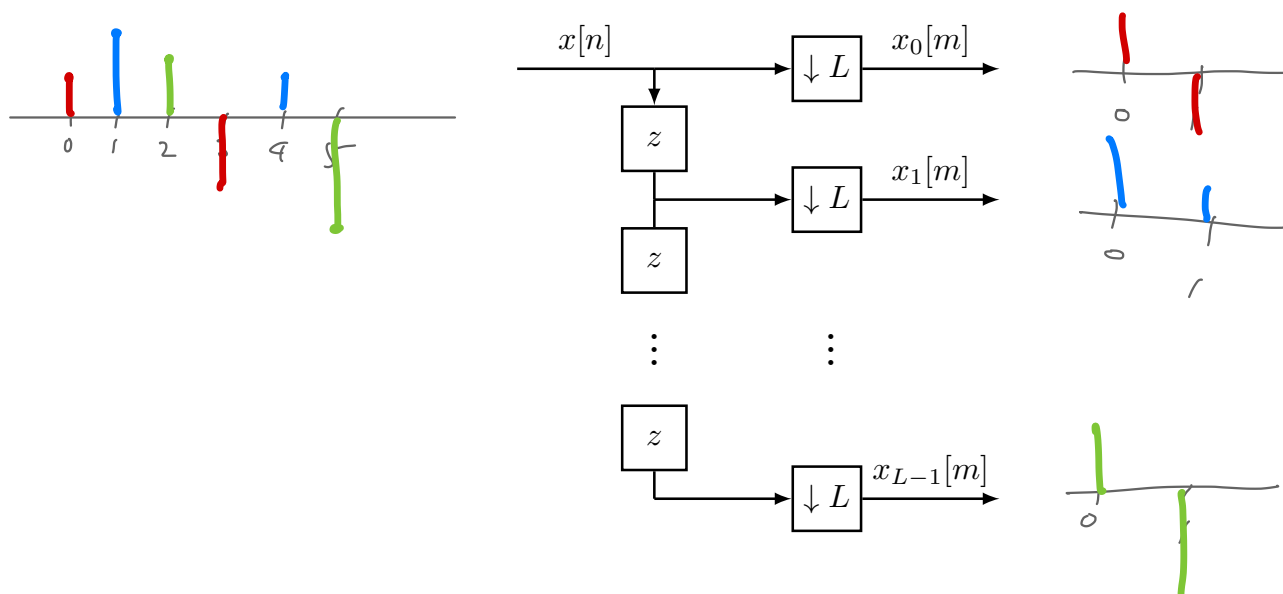


Polyphase Decomposition:

Polyphase components

$$x_k[m] = x[Lm + k], \quad k = 0, 1, \dots, L - 1$$

are obtained by the following block diagram:



Polyphase Decomposition of Nyquist-Rate Samples:

Let $(d_0, d_1, \dots, d_{L-1})$ denote the selection sequence such that

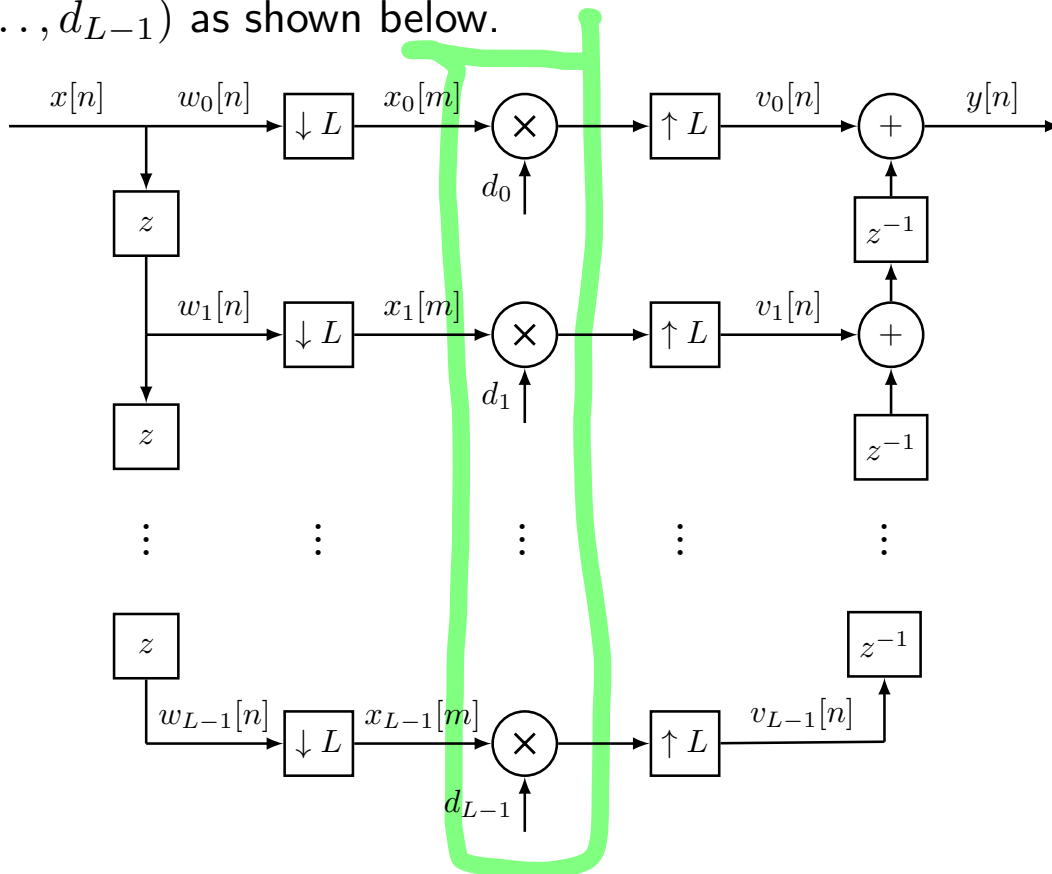
$$k \in \Phi \implies d_k = 1, \quad k \notin \Phi \implies d_k = 0.$$

Example: $L = 4, p = 3$

$$\Phi = \{0, 1, 3\} \iff \begin{bmatrix} d_0 \\ d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

*index set
of selected components*

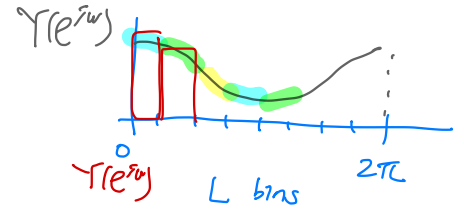
We consider a subset of polyphase components with respect to $(d_0, d_1, \dots, d_{L-1})$ as shown below.



$d_k = 1 \rightarrow$ select k th component / $d_k = 0 \rightarrow$ discard k th component

The output is written in the time domain as

$$y[n] = \sum_{k=0}^{L-1} v_k[n - k].$$



Therefore, we have

$$Y(e^{j\omega}) = \sum_{k=0}^{L-1} V_k(e^{j\omega}) e^{-j\omega k} = \sum_{k=0}^{L-1} d_k X_k(e^{j\omega L}) e^{-j\omega k}.$$

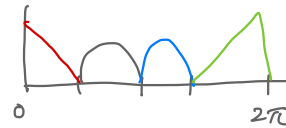
upsampling

By substituting ω by $\omega + \frac{2\pi i}{L}$, we obtain

$$Y(e^{j(\omega + \frac{2\pi i}{L})}) = \sum_{k=0}^{L-1} d_k X_k(e^{j\omega L}) e^{-j\omega k} e^{-j\frac{2\pi i k}{L}}, \quad i = 0, 1, \dots, L-1.$$

for $0 \leq \omega < \frac{2\pi}{L}$

The set of identities is rearranged to



$$\underbrace{\begin{bmatrix} Y(e^{j\omega}) \\ Y(e^{j(\omega + \frac{2\pi}{L})}) \\ \vdots \\ Y(e^{j(\omega + \frac{2\pi(L-1)}{L})}) \end{bmatrix}}_{\gamma(\omega)} = \mathbf{W} \underbrace{\begin{bmatrix} d_0 e^{-j0} X_0(e^{j\omega L}) \\ d_1 e^{-j\omega} X_1(e^{j\omega L}) \\ \vdots \\ d_{L-1} e^{-j\omega(L-1)} X_{L-1}(e^{j\omega L}) \end{bmatrix}}_{\alpha(\omega)},$$

where $\mathbf{W} \in \mathbb{C}^{L \times L}$ denotes the DFT matrix of size L . Let $\gamma_i(\omega)$ denotes the i th entry of $\gamma(\omega)$ for $i = 0, 1, \dots, L-1$, that is,

$$\gamma(\omega) = \begin{bmatrix} \gamma_0(\omega) \\ \gamma_1(\omega) \\ \vdots \\ \gamma_{L-1}(\omega) \end{bmatrix}.$$

Furthermore, by the polyphase decomposition, we also have

$$w_k[n] = x[n + k], \quad x_k[m] = w_k[Lm]$$

$$W_k(e^{j\omega}) = X(e^{j\omega})e^{j\omega k}, \quad X_k(e^{j\omega}) = \frac{1}{L} \sum_{l=0}^{L-1} W_k(e^{j(\omega - \frac{2\pi l}{L})}).$$

Then it follows that

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{k=0}^{L-1} V_k(e^{j\omega})e^{-j\omega k} = \sum_{k=0}^{L-1} d_k X_k(e^{j\omega L})e^{-j\omega k} \\ &= \sum_{k=0}^{L-1} d_k \frac{1}{L} \sum_{l=0}^{L-1} X(e^{j(\omega - \frac{2\pi l}{L})})e^{j\omega k} e^{-j\frac{2\pi lk}{L}} e^{-j\omega k} \\ &= \sum_{k=0}^{L-1} d_k \frac{1}{L} \sum_{l=0}^{L-1} X(e^{j(\omega - \frac{2\pi l}{L})})e^{-j\frac{2\pi lk}{L}}. \end{aligned}$$

Again, by substituting ω by $\omega + \frac{2\pi i}{L}$, we obtain

$$\begin{aligned} \underbrace{Y(e^{j(\omega + \frac{2\pi i}{L})})}_{\gamma_i(\omega)} &= \sum_{k=0}^{L-1} d_k \frac{1}{L} \sum_{l=0}^{L-1} X(e^{j(\omega + \frac{2\pi(i-l)}{L})})e^{-j\frac{2\pi lk}{L}} \\ &= \frac{1}{L} \sum_{l=0}^{L-1} \underbrace{\sum_{k=0}^{L-1} d_k e^{-j\frac{2\pi lk}{L}}}_{g_l} \underbrace{X(e^{j(\omega + \frac{2\pi(i-l)}{L})})}_{\beta_{i-l}(\omega)}, \end{aligned}$$

which is rewritten as a circular convolution as follows:

$$\gamma_i(\omega) = \frac{1}{L} \sum_{l=0}^{L-1} g_l \beta_{i-l}(\omega).$$

circular convolution
of g_l and β_l at i

g_l as function of l
is L -periodic

β_l is also L -periodic
function of l

By collecting the equation for all $i = 0, 1, \dots, L - 1$, we obtain

$$\underbrace{\begin{bmatrix} \gamma_0(\omega) \\ \gamma_1(\omega) \\ \gamma_2(\omega) \\ \vdots \\ \gamma_{L-1}(\omega) \end{bmatrix}}_{\boldsymbol{\gamma}(\omega)} = \frac{1}{L} \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ \vdots \\ g_{L-1} \end{bmatrix} \circledast \underbrace{\begin{bmatrix} \beta_0(\omega) \\ \beta_1(\omega) \\ \beta_2(\omega) \\ \vdots \\ \beta_{L-1}(\omega) \end{bmatrix}}_{\boldsymbol{\beta}(\omega)}.$$

The set of equations

$$g_l = \sum_{k=0}^{L-1} d_k e^{-j \frac{2\pi l k}{L}}, \quad l = 0, 1, \dots, L - 1$$

is written as

$$\underbrace{\begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ \vdots \\ g_{L-1} \end{bmatrix}}_{\mathbf{g}} = \mathbf{W} \underbrace{\begin{bmatrix} d_0 \\ d_1 \\ d_2 \\ \vdots \\ d_{L-1} \end{bmatrix}}_{\mathbf{d}}.$$

We have shown

$$\boldsymbol{\gamma}(\omega) = \frac{1}{L} \mathbf{g} \circledast \boldsymbol{\beta}(\omega).$$

By applying the inverse DFT to both sides, we obtain

$$\begin{aligned}\mathbf{W}^{-1}\boldsymbol{\gamma}(\omega) &= \mathbf{W}^{-1}\mathbf{g} \odot \mathbf{W}^{-1}\boldsymbol{\beta}(\omega) \\ &= \mathbf{d} \odot \mathbf{W}^{-1}\boldsymbol{\beta}(\omega) \\ &= \text{diag}(\mathbf{d})\mathbf{W}^{-1}\boldsymbol{\beta}(\omega) \\ &= \begin{bmatrix} d_0 & & & \\ & d_1 & & \\ & & \ddots & \\ & & & d_{L-1} \end{bmatrix} \mathbf{W}^{-1}\boldsymbol{\beta}(\omega).\end{aligned}$$

Since $\boldsymbol{\gamma}(\omega) = \mathbf{W}\boldsymbol{\alpha}(\omega)$ or equivalently $\mathbf{W}^{-1}\boldsymbol{\gamma}(\omega) = \boldsymbol{\alpha}(\omega)$, we have

$$\boldsymbol{\alpha}(\omega) = \begin{bmatrix} d_0 & & & \\ & d_1 & & \\ & & \ddots & \\ & & & d_{L-1} \end{bmatrix} \mathbf{W}^{-1}\boldsymbol{\beta}(\omega).$$

Selection Matrices:

We consider selecting p polyphase components out of L . Then

$$\Phi = \{k \in \{0, 1, 2, \dots, L-1\} : d_k = 1\} = \{\phi_1, \phi_2 \dots \phi_p\}.$$

Let $\mathbf{E}_\Phi^\top \in \mathbb{R}^{p \times L}$ denote the **selection matrix** such that

$$\mathbf{E}_\Phi^\top \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{L-1} \end{bmatrix} = \begin{bmatrix} x_{\phi_1} \\ x_{\phi_2} \\ \vdots \\ x_{\phi_p} \end{bmatrix}.$$

$$\mathbf{E}_\Phi^\top \mathbf{E}_\Phi = \mathbf{I}_p$$

Then we have

$$\mathbf{E}_\Phi \mathbf{E}_\Phi^\top = \begin{bmatrix} d_0 & & & \\ & d_1 & & \\ & & \ddots & \\ & & & d_{L-1} \end{bmatrix}.$$

$$\phi_1 = 0 \quad \phi_2 = 1 \quad \phi_3 = 3$$

Example: $L = 4$, $p = 3$, $\Phi = \{0, 1, 3\}$, $d_0 = 1$, $d_1 = 1$, $d_2 = 0$, $d_3 = 1$

The corresponding selection matrix is written as

$$\mathbf{E}_\Phi^\top = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} a_0 \\ a_1 \\ a_3 \end{bmatrix}$$

$$\mathbf{E}_\Phi \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

and satisfies

$$\mathbf{E}_\Phi = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{E}_\Phi \mathbf{E}_\Phi^\top = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 0 & \\ & & & 1 \end{bmatrix}.$$

$$\mathbf{E}_\Phi^\top \mathbf{E}_\Phi \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \mathbf{E}_\Phi^\top \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Simplification with Selection Matrices:

We have shown that

$$\boldsymbol{\alpha}(\omega) = \begin{bmatrix} d_0 & & & \\ & d_1 & & \\ & & \ddots & \\ & & & d_{L-1} \end{bmatrix} \mathbf{W}^{-1} \boldsymbol{\beta}(\omega),$$

where

$$\boldsymbol{\alpha}(\omega) = \begin{bmatrix} d_0 e^{-j0} X_0(e^{j\omega L}) \\ d_1 e^{-j\omega} X_1(e^{j\omega L}) \\ \vdots \\ d_{L-1} e^{-j\omega(L-1)} X_{L-1}(e^{j\omega L}) \end{bmatrix}, \quad \boldsymbol{\beta}(\omega) = \begin{bmatrix} X(e^{j\omega}) \\ X(e^{j(\omega + \frac{2\pi}{L})}) \\ \vdots \\ X(e^{j(\omega + \frac{2\pi(L-1)}{L})}) \end{bmatrix}.$$

"observed" polyphase components
DFT of Nyquist-sampled signal.

Since

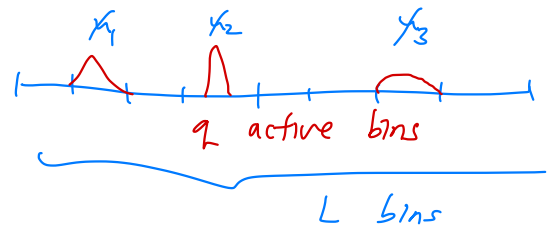
$$\mathbf{E}_\Phi \mathbf{E}_\Phi^\top = \begin{bmatrix} d_0 & & & \\ & d_1 & & \\ & & \ddots & \\ & & & d_{L-1} \end{bmatrix},$$

we obtain

$$\underbrace{\mathbf{E}_\Phi^\top}_{\checkmark} \boldsymbol{\alpha}(\omega) = \underbrace{\mathbf{E}_\Phi^\top}_{\checkmark} \mathbf{E}_\Phi \mathbf{E}_\Phi^\top \mathbf{W}^{-1} \boldsymbol{\beta}(\omega).$$

Furthermore, since $\mathbf{E}_\Phi^\top \mathbf{E}_\Phi = \mathbf{I}_p$, we have

$$\underbrace{\mathbf{E}_\Phi^\top \boldsymbol{\alpha}(\omega)}_{\text{observed polyphase components}} = \mathbf{E}_\Phi^\top \mathbf{W}^{-1} \underbrace{\boldsymbol{\beta}(\omega)}_{\text{desired Nyquist-sampled signal}}.$$



Plugging in Multiband Model:

Remember that $X(e^{j\omega})$ is sparse in the sense that it has at most q active subbands indexed by Ψ out of L spectral bins. Let $\Psi = \{\psi_1, \psi_2, \dots, \psi_q\} \subset \{0, 1, \dots, L-1\}$. Then similarly we can construct a selection matrix $\mathbf{E}_\Psi$ such that

$$\mathbf{E}_\Psi^\top \mathbf{E}_\Psi = \mathbf{I}_q$$

$$\mathbf{E}_\Psi \mathbf{E}_\Psi^\top \boldsymbol{\beta}(\omega) = \boldsymbol{\beta}(\omega).$$

By plugging in this to the previous identity

$$\mathbf{E}_\Phi^\top \boldsymbol{\alpha}(\omega) = \mathbf{E}_\Phi^\top \mathbf{W}^{-1} \boldsymbol{\beta}(\omega),$$

we obtain

$$\mathbf{E}_\Phi^\top \boldsymbol{\alpha}(\omega) = \mathbf{E}_\Phi^\top \mathbf{W}^{-1} \mathbf{E}_\Psi \mathbf{E}_\Psi^\top \boldsymbol{\beta}(\omega).$$

Ex L : prime number.

$\Phi = \{0, 1, 2, \dots, p-1\}$

$p > q$

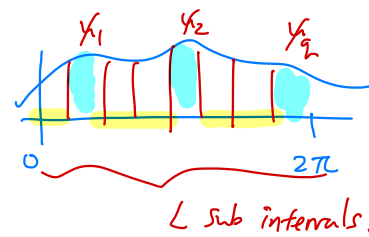
$\mathbf{E}_\Phi^\top \mathbf{W}^{-1} \mathbf{E}_\Psi$ has full column rank.

Let $\mathbf{A} \triangleq \mathbf{E}_\Phi^\top \mathbf{W}^{-1} \mathbf{E}_\Psi \in \mathbb{C}^{p \times q}$, which is a submatrix of the inverse DFT matrix of size L obtained by choosing rows by Φ and columns by Ψ .

If $\mathbf{A}$ has full column rank, there exists $\mathbf{A}^\dagger \in \mathbb{C}^{q \times p}$ such that $\mathbf{A}^\dagger \mathbf{A} = \mathbf{I}_q$. Then we have

$$\underbrace{\mathbf{E}_\Psi^\top \boldsymbol{\beta}(\omega)}_{\text{Desired}} = \mathbf{A}^\dagger \underbrace{\mathbf{E}_\Phi^\top \boldsymbol{\alpha}(\omega)}_{\text{Observed}}.$$

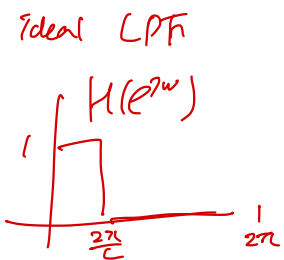
There exist universally good designs of decimation pattern Φ so that $\mathbf{A} = \mathbf{E}_\Phi^\top \mathbf{W}^{-1} \mathbf{E}_\Psi$ has full column rank for all Ψ with $|\Psi| = q < p$. Furthermore, the support index set Ψ can be estimated from cross correlations of polyphase signals.



Reconstruction System:

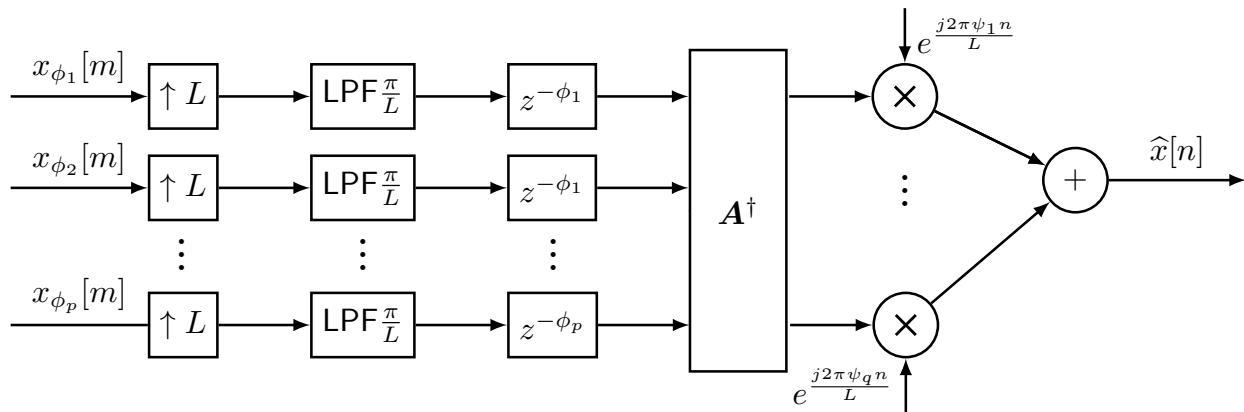
The Nyquist-rate signal $x[n]$ can be reconstructed by

$$\begin{aligned}
 x[n] &= \frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \\
 &= \frac{1}{2\pi} \int_0^{\frac{2\pi}{L}} \sum_{i=0}^{q-1} X(e^{j(\omega + \frac{2\pi\psi_i}{L})}) e^{j(\omega + \frac{2\pi\psi_i}{L})n} d\omega \\
 &= \sum_{i=0}^{q-1} e^{j\frac{2\pi\psi_i}{L}n} \underbrace{\frac{1}{2\pi} \int_0^{\frac{2\pi}{L}} X(e^{j(\omega + \frac{2\pi\psi_i}{L})}) e^{j\omega n} d\omega}_{\text{inverse DTFT of } H(e^{j\omega}) X(e^{j(\omega + \frac{2\pi\psi_i}{L})})}
 \end{aligned}$$



inverse DTFT of $H(e^{j\omega}) X(e^{j(\omega + \frac{2\pi\psi_i}{L})})$

The reconstruction system is described by the following block diagram.



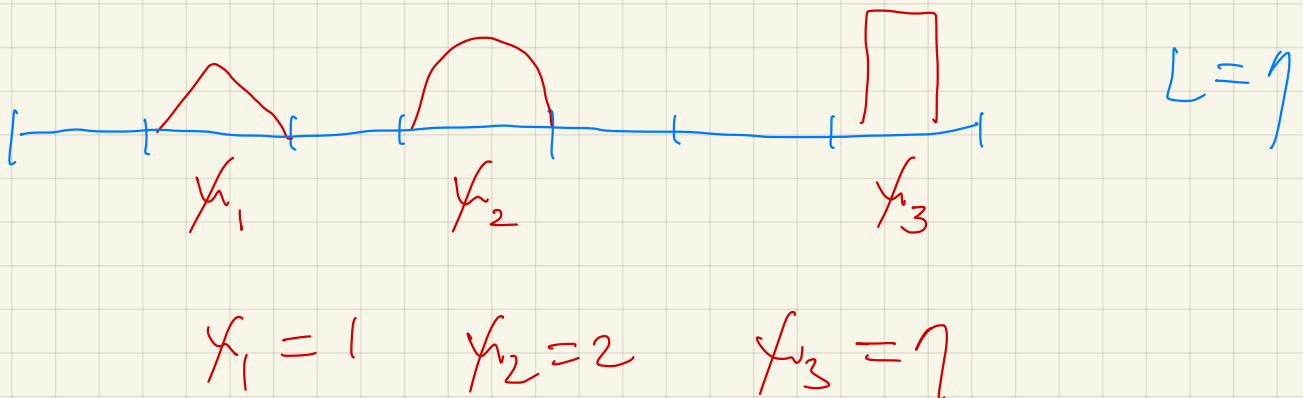
$$\binom{L}{q} = \frac{L!}{q!(L-q)!} \approx \left(\frac{Le}{q}\right)^q$$

$$\underbrace{E_{\Phi}^T}_{\text{red}} \alpha(\omega) = \overbrace{E_{\Phi}^T W^{-1} E_{\Phi}}^A E_{\Phi}^T \beta(\omega)$$

$$y(\omega) = \begin{bmatrix} e^{-j\phi_1 \omega} X_{\phi_1}(e^{j\omega L}) \\ e^{-j\phi_2 \omega} X_{\phi_2}(e^{j\omega L}) \\ \vdots \\ e^{-j\phi_p \omega} X_{\phi_p}(e^{j\omega L}) \end{bmatrix}$$

$$s(\omega) = E_{\Psi}^T \beta(\omega) = \begin{bmatrix} x(e^{j(\omega + \frac{2\pi\psi_1}{L})}) \\ x(e^{j(\omega + \frac{2\pi\psi_2}{L})}) \\ \vdots \\ x(e^{j(\omega + \frac{2\pi\psi_q}{L})}) \end{bmatrix}$$

ex)



Define

$$R_y \triangleq \int_0^{\frac{2\pi}{L}} y(\omega) y(\omega)^H d\omega$$

$$= A \underbrace{\int_0^{\frac{2\pi}{L}} s(\omega) s(\omega)^H d\omega}_{R_s \in \mathbb{C}^{q \times q}} A^H$$

Assume R_s is full rank.

$$\text{Let } B = E_{\mathbb{I}}^T W^{-1} \in \mathbb{C}^{p \times L}$$

A is a submatrix of B

with columns indexed by $\mathbb{I}$

$$(A = B E_{\mathbb{I}})$$

$$R_y = A R_s A^H \quad \& \quad R_s \text{ full rank}$$

$$\Rightarrow \text{span}(R_y) = \text{span}(A)$$

Suppose that any $q+1$ columns of B are linearly independent.

Let $Q \in \mathbb{C}^{p \times q}$ satisfy

$$Q^H Q = I_{p-q}$$

$$\text{span}(Q) = \text{span}(R_y)^\perp$$

Then the k -th column of B satisfies.

$$Q^H b_k = 0 \quad \text{iff} \quad k \in \underline{q}$$

(MUSIC)

How to calculate R_y ?

$$(R_y)_{k,l} = \int_0^{\frac{2\pi}{L}} e^{-j\phi_k \omega} X_{\phi_k}(e^{j\omega}) e^{j\phi_l \omega} X_{\phi_l}(e^{j\omega}) d\omega$$

